

9.7 Differentiating in Polar Form

Calculus

Name: _____

CA #1

1. Find the slope of the tangent line to the polar curve $r = 2 + 4 \sin \theta$ at $\theta = \pi$.
2. A particle moves along the polar curve $r = 4 - 2 \cos \theta$ so that $\frac{d\theta}{dt} = 4$. Find the value of $\frac{dr}{dt}$ at $\theta = \frac{\pi}{3}$.
3. For $0 \leq \theta \leq \pi$, find the values of θ for which the polar curve $r = 3 \sin \theta$ **might** have a vertical tangent line. Then use a graphing utility to eliminate any of your possible answers.
4. A polar curve is given by the equation $r = 2 \csc \theta + 3$ for $\theta \geq 0$. What is the instantaneous rate of change of r with respect to θ where $\theta = \frac{\pi}{4}$.
5. **Calculator active.** For a certain polar curve $r = f(\theta)$, it is known that $\frac{dx}{d\theta} = 3 \cos \theta - 3\theta \sin \theta$ and $\frac{dy}{d\theta} = 3(\sin \theta + \theta \cos \theta)$. What is the value of $\frac{d^2y}{dx^2}$ at $\theta = 3$?

1. $-\frac{2}{1}$	2. $4\sqrt{3}$	3. $\frac{\pi}{3}, \frac{\pi}{4}$	4. $-2\sqrt{2}$	5. -1.299
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